

PHOENIX DEER VALLEY, AZ, USA

WMO: 722784

Lat: 33.688N

Lon: 112.082W

Elev: 1455

StdP: 13.94

Time Zone: -7.00 (W07)

Period: 98-19

WBAN: 03184

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	36.6	39.3	-1.0	5.5	60.0	3.3	6.9	61.7	19.5	54.6	17.0	55.8	3.2	20	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	108.3	68.9	106.3	68.5	104.2	68.0	74.7	92.8	73.8	92.2	73.0	92.2	10.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	120.7	81.0	69.1	113.1	81.8	66.4	102.7	84.6	39.1	92.8	38.3	91.7	37.5	91.6	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.3	13.6	DB	31.3	112.6	3.6	2.3	28.7	114.2	26.6	115.5	24.6	116.8	22.0	118.4	(4)
(5)			WB	25.5	76.5	2.9	2.8	23.4	78.5	21.7	80.1	20.1	81.7	18.0	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.0	55.2	57.2	63.6	69.9	78.5	88.6	92.8	91.3	86.3	74.8	63.2	53.8		
	DBStd	15.11	6.02	6.18	6.73	6.71	6.74	5.10	4.28	4.30	5.17	6.55	7.19	5.79		
	HDD50	65	22	11	1	0	0	0	0	0	0	0	2	29		
	HDD65	1142	306	223	107	30	3	0	0	0	0	9	117	347		
	CDD50	8469	183	214	424	597	882	1158	1328	1280	1090	768	398	147		
	CDD65	4072	2	6	65	177	420	708	863	815	640	312	63	1		
	CDH74	60975	33	107	798	2134	5814	11623	14246	13069	9269	3303	567	12		
(13)	CDH80	37924	0	6	213	852	3225	7807	9873	8742	5603	1467	136	0		
(14)	Wind	WSAvg	5.7	4.5	4.9	5.7	6.8	6.5	6.9	6.2	5.9	5.2	4.6	4.3		
(15) Precipitation	PrecAvg	7.9	0.9	0.8	0.7	0.4	0.1	0.1	0.7	1.3	0.7	0.6	0.5	1.0		
	PrecMax	11.7	3.7	4.2	1.8	1.5	1.2	0.4	1.8	2.6	3.7	3.9	1.9	2.2		
	PrecMin	4.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.3	0.0	0.0	0.0	0.2		
	PrecStd	2.3	1.1	1.2	0.6	0.4	0.3	0.1	0.5	0.7	0.9	0.8	0.5	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.1	81.0	90.3	95.2	104.5	111.1	111.0	109.3	105.7	97.8	88.4	75.7		
		MCWB	51.7	53.4	57.9	58.9	65.7	66.4	70.3	70.6	68.7	64.3	58.4	52.5		
	2%	DB	73.5	77.5	85.1	91.4	100.0	107.6	108.6	106.9	102.4	93.7	83.8	72.0		
		MCWB	50.7	52.4	55.6	57.9	63.3	65.7	69.9	70.1	67.9	62.8	56.3	51.4		
	5%	DB	70.5	73.9	81.9	88.4	97.3	105.6	106.4	104.7	100.4	91.1	80.6	68.7		
		MCWB	49.7	50.9	55.0	57.0	61.3	65.7	69.7	69.8	67.2	61.6	55.5	50.1		
	10%	DB	67.2	70.4	78.8	85.1	94.1	102.6	104.0	102.1	98.7	88.4	76.8	65.3		
		MCWB	48.4	50.4	53.4	55.4	60.0	64.4	69.3	69.7	67.0	60.7	54.0	48.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	57.8	59.5	64.2	69.1	73.0	76.4	77.9	74.3	70.5	62.1	57.7		
		MCDB	65.0	68.3	82.5	87.5	100.1	99.6	95.8	97.7	90.7	80.2	74.0	64.9		
	2%	WB	54.6	55.9	58.0	60.1	66.1	71.1	74.8	74.9	72.8	67.1	59.2	55.0		
		MCDB	64.2	67.0	79.8	86.5	94.0	97.0	93.1	92.7	88.9	86.6	75.6	65.7		
	5%	WB	52.8	54.1	56.4	58.2	63.8	69.4	73.9	73.9	71.6	64.9	57.2	52.8		
		MCDB	63.8	66.5	76.8	83.6	91.3	96.3	92.5	92.2	89.6	84.6	75.7	63.5		
	10%	WB	50.9	52.4	54.8	56.7	61.4	67.6	73.1	73.1	70.4	62.9	55.4	50.7		
		MCDB	63.2	66.3	74.2	81.6	90.8	97.3	92.5	92.1	89.8	82.6	73.7	62.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.6	20.8	23.3	24.1	24.9	25.8	20.7	20.0	20.9	21.7	21.2	19.7		
		MCDBR	24.9	26.4	27.8	28.0	28.3	28.0	23.0	22.2	22.7	24.1	24.9	23.5		
		MCWBR	11.9	12.2	10.8	11.7	11.2	10.3	7.6	6.7	8.2	10.2	11.3	11.4		
	5% WB	MCDBR	18.2	20.7	25.4	25.4	26.0	22.7	19.4	18.8	18.3	19.7	20.8	18.5		
		MCWBR	9.5	9.9	10.1	11.3	11.4	9.7	5.8	5.8	7.0	9.2	10.4	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.284	0.310	0.321	0.331	0.339	0.400	0.387	0.343	0.318	0.287	0.279		
	taud		2.582	2.580	2.484	2.438	2.445	2.420	2.378	2.402	2.500	2.522	2.564	2.588		
	Ebn at Noon		301	310	307	306	302	298	279	282	292	293	295	295		
	Edh at Noon		25	27	33	36	36	37	38	37	32	29	25	23		
(44) All-Sky Solar Radiation	RadAvg		1063	1364	1839	2293	2574	2680	2324	2140	1929	1550	1182	961		
	RadStd		82	109	94	70	96	101	111	76	72	93	51	53		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-198	N/A	N/A	
(47)	Regional (13 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-148	+193	+72	

Nomenclature: See separate page